

Activity using SQLite

DSA05 – Query Processing for Data Science

S. Sharon Surya

192324246

1. Create the Department Table (Parent Table) with attributes Department_ID INTEGER PRIMARY KEY, Department_Name TEXT

- Creates the Department table in the SQLite database.
- Defines Department_ID as the Primary Key to uniquely identify each department.
- Stores the department name in the Department_Name column.
- Saves the table permanently in the database using commit().

Program:

```
import sqlite3
```

```
# Connect to SQLite database
```

```
conn = sqlite3.connect("college.db")
```

```
# Create cursor
```

```
cursor = conn.cursor()
```

```
# Create Department table
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Department(
```

```
    Department_ID INTEGER PRIMARY KEY,
```

```
    Department_Name TEXT
```

```
)
```

```
""")
```

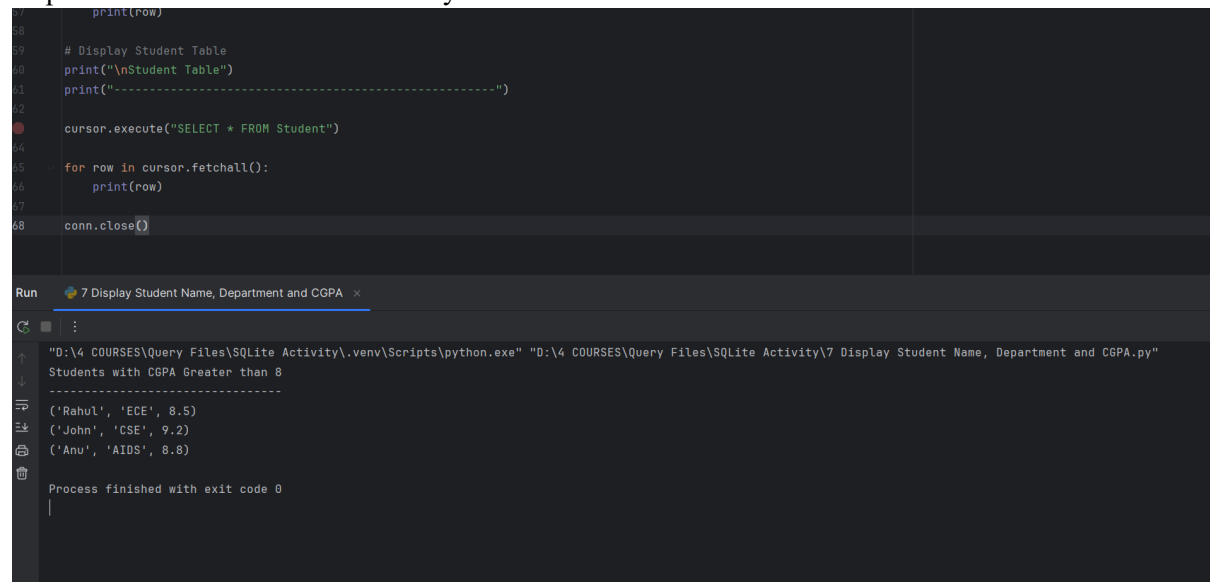
```
# Save changes
```

```
conn.commit()
```

```
print("Department table created successfully.")
```

OUTPUT:

Department table created successfully.



The screenshot shows a Python IDE with a dark theme. The editor window contains the following code:

```
67     print(row)
68
69 # Display Student Table
70 print("\nStudent Table")
71 print("-----")
72
73 cursor.execute("SELECT * FROM Student")
74
75 for row in cursor.fetchall():
76     print(row)
77
78 conn.close()
```

Below the editor is a 'Run' console window. It shows the command executed: `"D:\4 COURSES\Query Files\SQLite Activity\venv\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\7 Display Student Name, Department and CGPA.py"`. The output is:

```
Students with CGPA Greater than 8
-----
('Rahul', 'ECE', 8.5)
('John', 'CSE', 9.2)
('Anu', 'AIOS', 8.8)
```

At the bottom of the console, it says: `Process finished with exit code 0`.

2. Create the Student Table (Child Table)

- Creates the Student table to store student information.
- Defines Student_ID as the Primary Key for unique student identification.
- Uses Department_ID as a Foreign Key to connect with the Department table.
- Stores student details such as Name, Age, Department ID, and CGPA.

Program:

```
import sqlite3
```

```
# Connect to database
```

```
conn = sqlite3.connect("college.db")
```

```
cursor = conn.cursor()
```

```
# Create Department Table
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Department(
```

```
    Department_ID INTEGER PRIMARY KEY,
```

```
    Department_Name TEXT
```

```

)
)

# Create Student Table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Age INTEGER,
    Department_ID INTEGER,
    CGPA REAL,
    FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)
)
)

# Delete old data (to avoid duplicate entries when running again)
cursor.execute("DELETE FROM Student")
cursor.execute("DELETE FROM Department")

# Insert Departments
cursor.executemany("INSERT INTO Department VALUES (?,?)",[
    (1,"CSE"),
    (2,"ECE"),
    (3,"AIDS"),
    (4,"MECH")
])

# Insert Students
cursor.executemany("INSERT INTO Student VALUES (?,?,,?,?)",[
    (101,"Rahul",20,1,8.5),
    (102,"Priya",19,2,7.8),
    (103,"John",21,1,9.2),
    (104,"Anu",20,3,8.8),
    (105,"Kiran",22,4,7.5)
])

conn.commit()

# Display Department Table
print("\nDepartment Table")
print("-----")

cursor.execute("SELECT * FROM Department")

for row in cursor.fetchall():
    print(row)

# Display Student Table
print("\nStudent Table")
print("-----")

```

```
cursor.execute("SELECT * FROM Student")
```

```
for row in cursor.fetchall():  
    print(row)
```

```
conn.close()
```

OUTPUT:

('Rahul', 'ECE', 8.5)

('John', 'CSE', 9.2)

('Anu', 'AIDS', 8.8)

```
57     print(row)
58
59     # Display Student Table
60     print("\nStudent Table")
61     print("-----")
62
63     cursor.execute("SELECT * FROM Student")
64
65     for row in cursor.fetchall():
66         print(row)
67
68     conn.close()
```

Run 7 Display Student Name, Department and CGPA x

"D:\4 COURSES\Query Files\SQLite Activity\.venv\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\7 Display Student Name, Department and CGPA.py"

Students with CGPA Greater than 8

('Rahul', 'ECE', 8.5)

('John', 'CSE', 9.2)

('Anu', 'AIDS', 8.8)

Process finished with exit code 0

3: Display Students from CSE Department

- Retrieves student records belonging to the CSE department.
- Uses the **JOIN** operation to combine Student and Department tables.
- Filters the records using the **WHERE** clause.
- Displays the names of students who belong to the CSE department.

Program:

```
import sqlite3
```

```
conn = sqlite3.connect("college.db")
```

```
cursor = conn.cursor()
```

```

print("Students from CSE Department")
print("-----")

cursor.execute("""
SELECT Student.Name, Department.Department_Name
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Department.Department_Name = 'CSE'
""")

for row in cursor.fetchall():
    print(row)

conn.close()

import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

print("Students from CSE Department")
print("-----")

cursor.execute("""
SELECT Student.Name, Department.Department_Name
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Department.Department_Name = 'CSE'
""")

for row in cursor.fetchall():
    print(row)

conn.close()

```

OUTPUT:

Students from CSE Department

(John', 'CSE')

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

print("Students from CSE Department")
print("-----")

cursor.execute("""
SELECT Student.Name, Department.Department_Name
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Department.Department_Name = 'CSE'
""")
```

Display Students from CSE Department

```
"D:\4 COURSES\Query Files\SQLite Activity\.env\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\Display Students from CSE Department"
Students from CSE Department
-----
('John', 'CSE')

Process finished with exit code 0
```

4. Count Students in Each Department

- Counts the total number of students in every department.
- Uses the COUNT() aggregate function to calculate student totals.
- Groups the records department-wise using the GROUP BY clause.
- Displays the department name along with the corresponding student count.

Program:

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

print("Number of Students in Each Department")
print("-----")

cursor.execute("""
SELECT Department.Department_Name,
COUNT(Student.Student_ID)
```

```
FROM Department
LEFT JOIN Student
ON Department.Department_ID = Student.Department_ID
GROUP BY Department.Department_Name
""")
```

```
for row in cursor.fetchall():
    print(row)
```

```
conn.close()
```

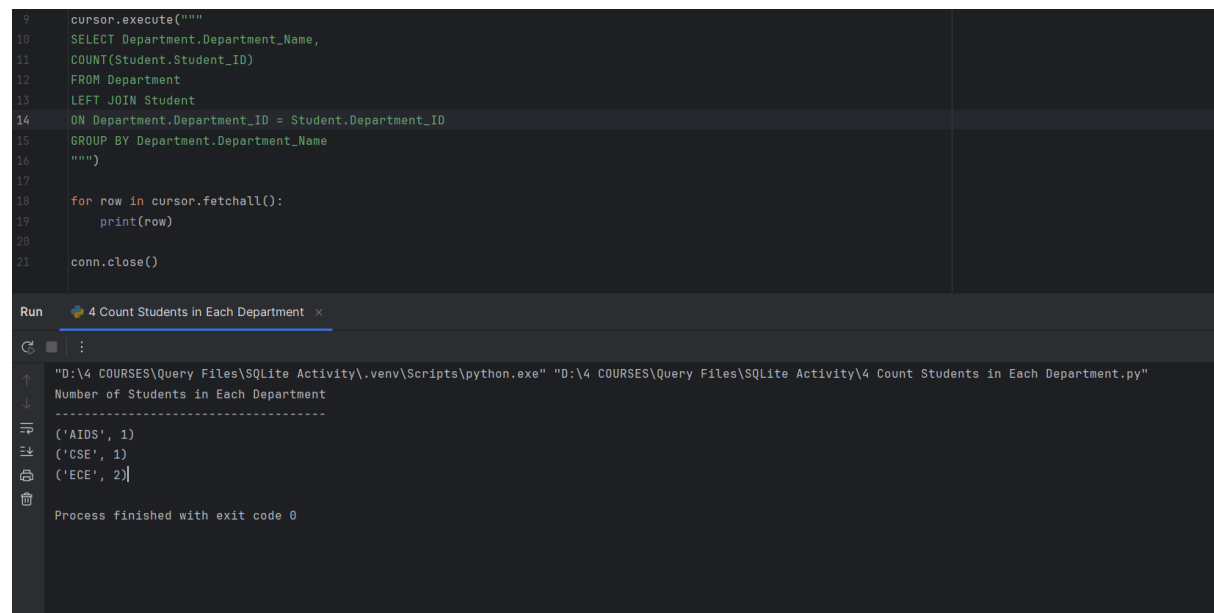
OUTPUT:

Number of Students in Each Department

('AIDS', 1)

('CSE', 1)

('ECE', 2)



The screenshot shows a Python IDE with a code editor and a run console. The code in the editor is as follows:

```
9 cursor.execute("""
10 SELECT Department.Department_Name,
11 COUNT(Student.Student_ID)
12 FROM Department
13 LEFT JOIN Student
14 ON Department.Department_ID = Student.Department_ID
15 GROUP BY Department.Department_Name
16 """)
17
18 for row in cursor.fetchall():
19     print(row)
20
21 conn.close()
```

The run console shows the output of the code:

```
Run 4 Count Students in Each Department x
"D:\4 COURSES\Query Files\SQLite Activity\.venv\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\4 Count Students in Each Department.py"
Number of Students in Each Department
-----
('AIDS', 1)
('CSE', 1)
('ECE', 2)
Process finished with exit code 0
```

5. Update the foreign key. For eg., change the department of a student

- Updates the Department_ID of a selected student.
- Changes the student's department without modifying other details.
- Uses the UPDATE statement with the WHERE clause.
- Displays the updated student record to verify the changes.

Program:

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
UPDATE Student
SET Department_ID = 2
WHERE Student_ID = 101
""")

conn.commit()

print("Updated Student Details")
print("-----")

cursor.execute("""
SELECT * FROM Student
WHERE Student_ID = 101
""")

for row in cursor.fetchall():
    print(row)

conn.close()
```

OUTPUT:

```
Updated Student Details
-----
(101, 'Rahul', 20, 2, 8.5)
```



```
7 cursor.execute("""
8 SELECT * FROM Student
9 WHERE Student_ID = 101
10 """)
11
12 for row in cursor.fetchall():
13     print(row)
14
15 conn.close()
```

Run 5 Update the Foreign Key (Change Department) x

```
"D:\4 COURSES\Query Files\SQLite Activity\.venv\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\5 Update the Foreign Key (Change Department).py"
Updated Student Details
-----
(101, 'Rahul', 20, 2, 8.5)
Process finished with exit code 0
```

6. Delete a department

- Deletes a selected department from the Department table.
- Removes related student records first to maintain foreign key integrity.
- Uses the DELETE statement to remove the department.
- Displays the remaining departments after deletion.

Program:

```
import sqlite3
```

```
conn = sqlite3.connect("college.db")
cursor = conn.cursor()
```

```
cursor.execute("DELETE FROM Student WHERE Department_ID = 4")
cursor.execute("DELETE FROM Department WHERE Department_ID = 4")
```

```
conn.commit()
```

```
print("Remaining Departments")
print("-----")
```

```
cursor.execute("SELECT * FROM Department")
```

```
for row in cursor.fetchall():
    print(row)
```

```
conn.close()
```

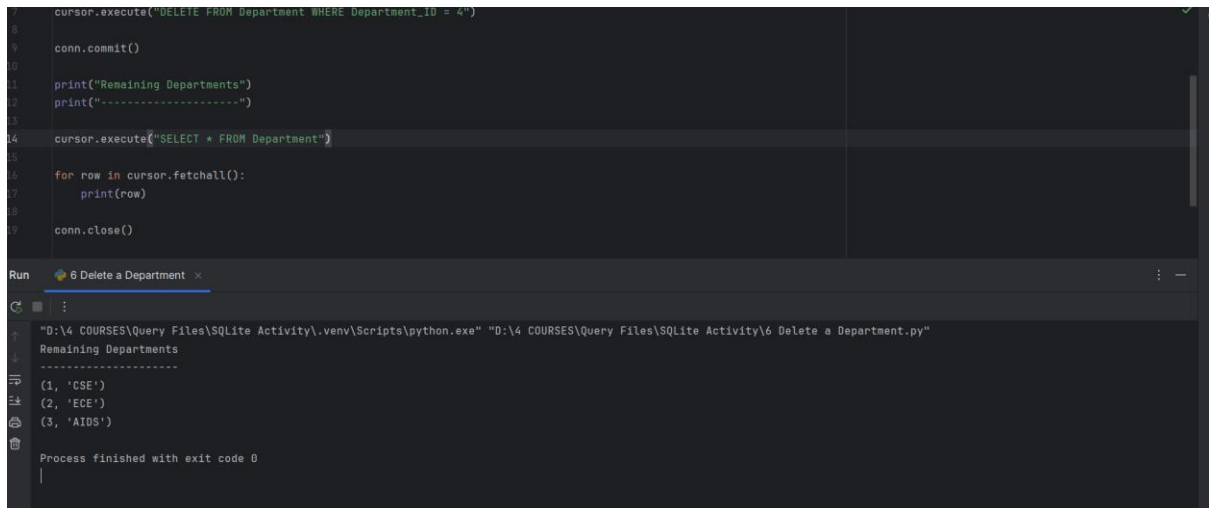
OUTPUT:

Remaining Departments

(1, 'CSE')

(2, 'ECE')

(3, 'AIDS')



```
1 cursor.execute("DELETE FROM Department WHERE Department_ID = 4")
2
3 conn.commit()
4
5 print("Remaining Departments")
6 print("-----")
7
8 cursor.execute("SELECT * FROM Department")
9
10 for row in cursor.fetchall():
11     print(row)
12
13 conn.close()
```

Run 6 Delete a Department x

```
"D:\4 COURSES\Query Files\SQLite Activity\venv\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\6 Delete a Department.py"
Remaining Departments
-----
(1, 'CSE')
(2, 'ECE')
(3, 'AIDS')
Process finished with exit code 0
```

7. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8.

- Retrieves students whose **CGPA is greater than 8**.
- Uses the **JOIN** operation to fetch both student and department details.
- Filters the records using the **WHERE** clause with the CGPA condition.
- Displays the student's name, department name, and CGPA

PROGRAM:

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

print("Students with CGPA Greater than 8")
print("-----")

cursor.execute("""
SELECT Student.Name,
Department.Department_Name,
Student.CGPA
```

```
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Student.CGPA > 8
""")
```

```
for row in cursor.fetchall():
    print(row)
```

```
conn.close()
```

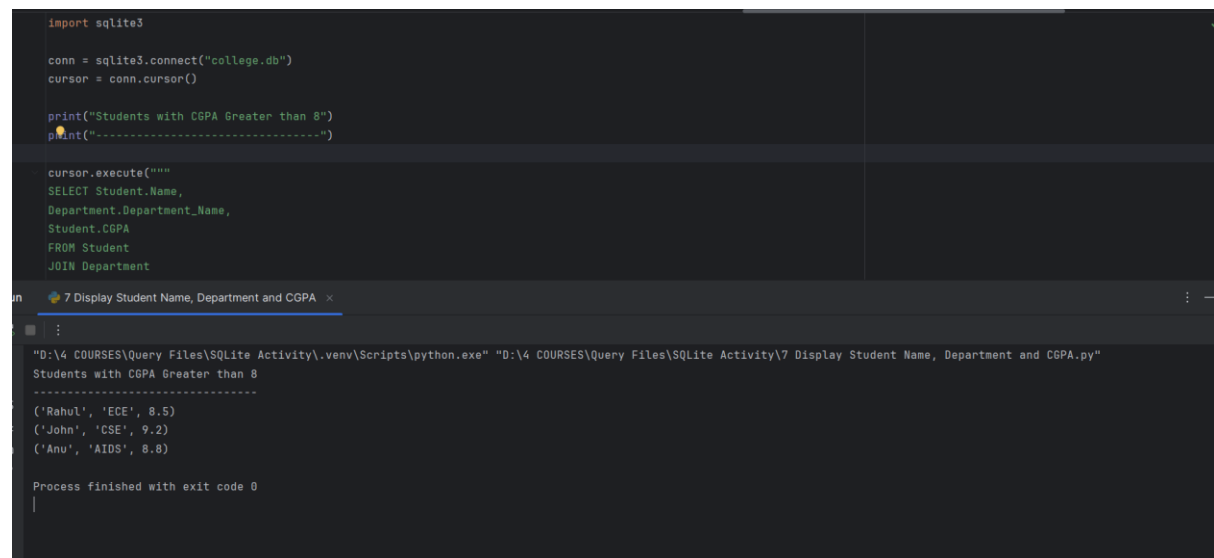
OUTPUT:

Students with CGPA Greater than 8

('Rahul', 'ECE', 8.5)

('John', 'CSE', 9.2)

('Anu', 'AIDS', 8.8)



```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

print("Students with CGPA Greater than 8")
print("-----")

cursor.execute("""
SELECT Student.Name,
Department.Department_Name,
Student.CGPA
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Student.CGPA > 8
""")

for row in cursor.fetchall():
    print(row)

conn.close()
```

7 Display Student Name, Department and CGPA

```
"D:\4 COURSES\Query Files\SQLite Activity\.venv\Scripts\python.exe" "D:\4 COURSES\Query Files\SQLite Activity\7 Display Student Name, Department and CGPA.py"
Students with CGPA Greater than 8
-----
('Rahul', 'ECE', 8.5)
('John', 'CSE', 9.2)
('Anu', 'AIDS', 8.8)

Process finished with exit code 0
```